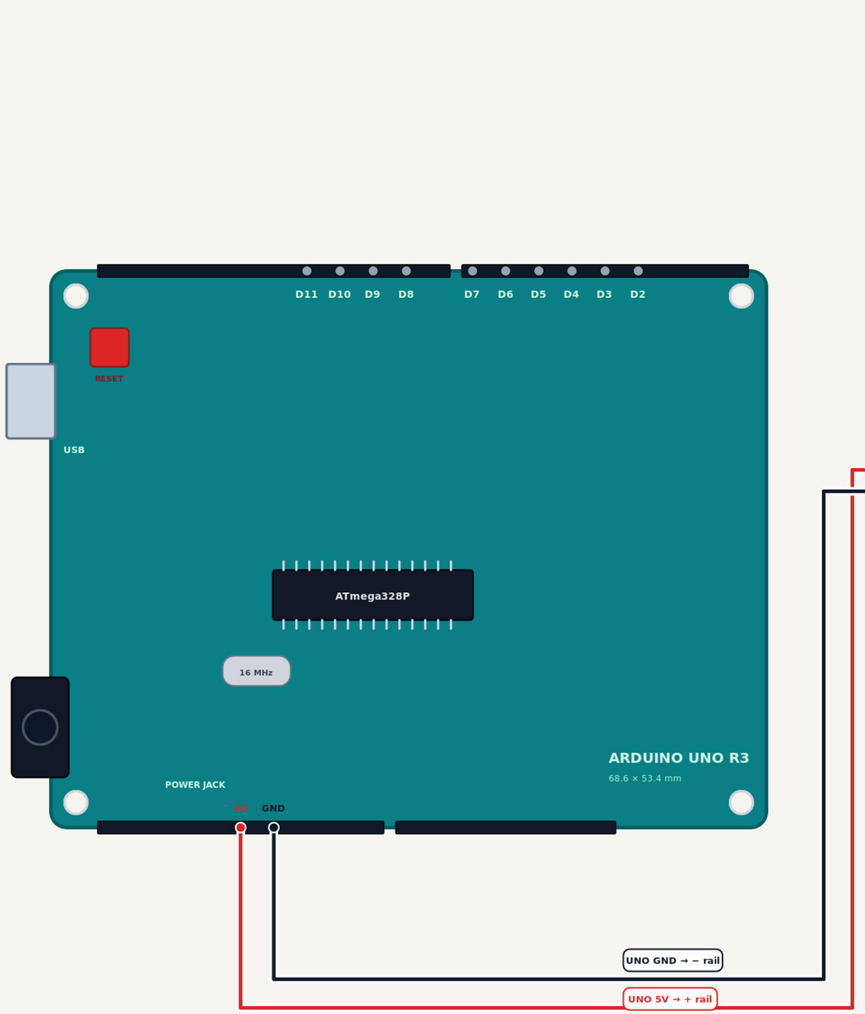
STEP 1 — POWER RAILS & COMMON GROUND

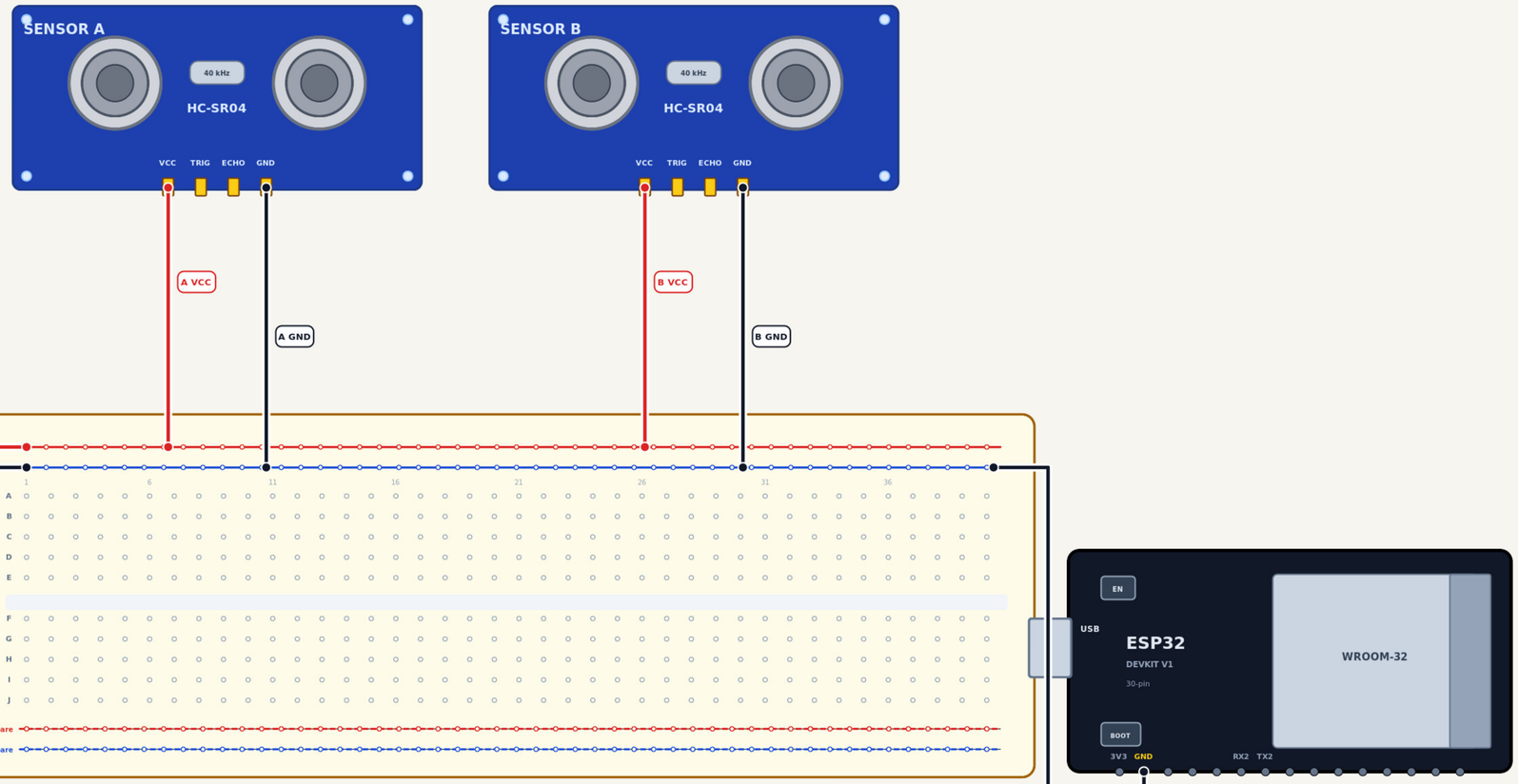
Connect 5 V, GND, sensor power and the ESP32 ground. No signal wires yet.

WIRING STEP
1 / 4

Keep USB disconnected while adding or moving wires. A dot at a wire end means a connection; a wire crossing another wire without a dot is NOT connected.
Only the components used in this step are shown. Complete the pages in order.



UNO 5V / GND are on the POWER header at the lower-left.



ESP32 is powered by its own USB cable; only GND is wired here.

CHECK BEFORE MOVING ON

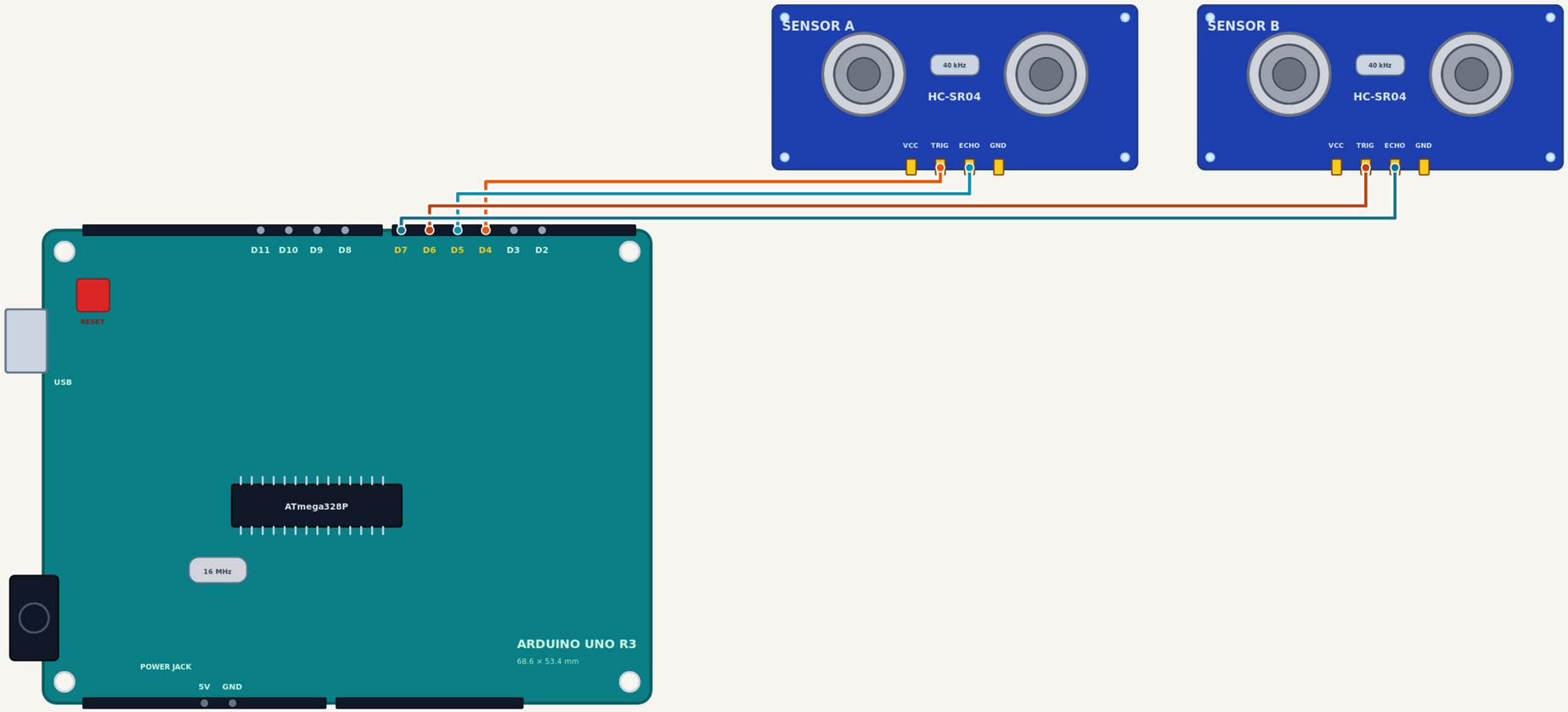
- UNO 5 V reaches the + rail and UNO GND reaches the − rail.
- Both HC-SR04 VCC pins are on +, both GND pins are on −.
- ESP32 GND is joined to the same − rail.
- No ESP32 5 V pin is connected.

STEP 2 — ULTRASONIC SENSOR SIGNALS

Four direct signal wires: sensor A/B TRIG and ECHO to UNO D4–D7.

WIRING STEP
2 / 4

Keep USB disconnected while adding or moving wires. A dot at a wire end means a connection; a wire crossing another wire without a dot is NOT connected.
Only the components used in this step are shown. Complete the pages in order.



Signal wires only. Sensor power is already on the rails from Step 1.

WIRE COLOUR KEY

- **orange = A TRIG → D4**
- **cyan = A ECHO → D5**
- **dark orange = B TRIG → D6**
- **dark cyan = B ECHO → D7**

CHECK BEFORE MOVING ON

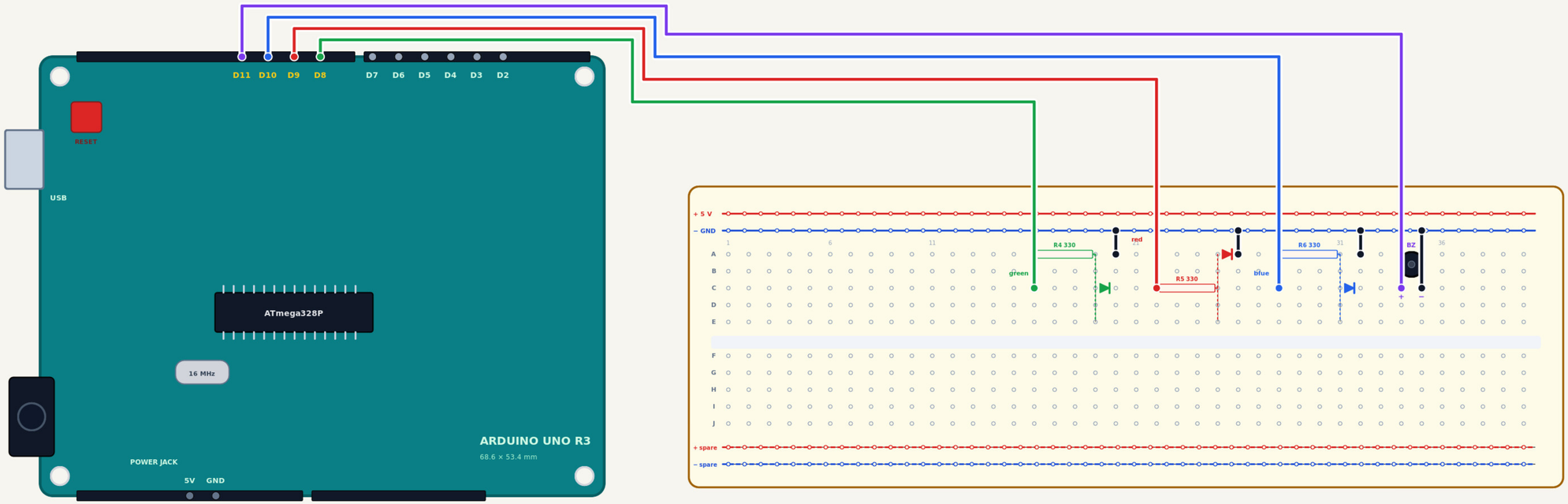
- Sensor A TRIG is on UNO D4; Sensor A ECHO is on UNO D5.
- Sensor B TRIG is on UNO D6; Sensor B ECHO is on UNO D7.
- Do not connect any HC-SR04 ECHO wire to the ESP32.
- Power wires from Step 1 remain in place.

STEP 3 — LEDS & ACTIVE BUZZER

Direct wires from UNO D8–D11 to the resistor inputs and buzzer; cathodes to GND.

WIRING STEP
3 / 4

Keep USB disconnected while adding or moving wires. A dot at a wire end means a connection; a wire crossing another wire without a dot is NOT connected.
Only the components used in this step are shown. Complete the pages in order.



Wires cross the top power rails in the drawing; they do not connect to them.

WIRE COLOUR KEY

green = D8 / green LED
red = D9 / red LED
blue = D10 / blue LED
purple = D11 / buzzer

Dotted vertical line = same breadboard column, already connected internally.

CHECK BEFORE MOVING ON

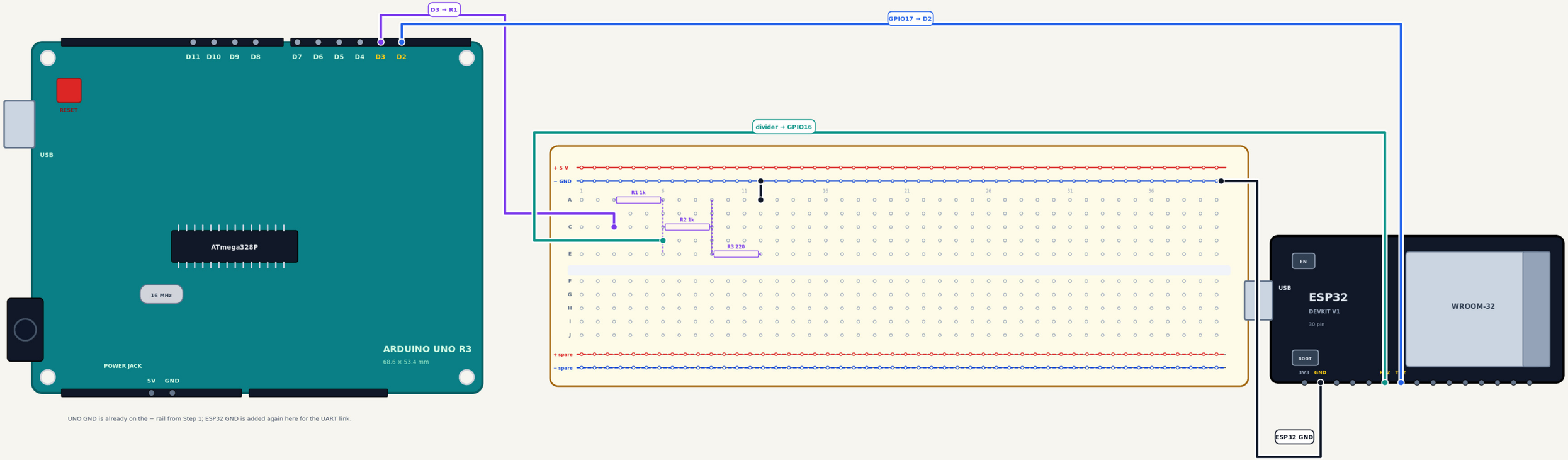
- R4, R5, R6 are 220–330 Ω, one per LED. These are still missing from your parts set.
- D8 → R4 → green LED; D9 → R5 → red LED; D10 → R6 → blue LED.
- Each LED cathode goes to the − rail. Long leg = anode, short/flat = cathode.
- D11 → buzzer +; buzzer − → − rail. Use an NPN driver if the buzzer draws more than 20 mA.

STEP 4 — UART DIVIDER & ESP32

Direct wires for UNO D3, the 5 V→3.3 V divider, ESP32 GPIO16/17 and common ground.

WIRING STEP
4 / 4

Keep USB disconnected while adding or moving wires. A dot at a wire end means a connection; a wire crossing another wire without a dot is NOT connected.
Only the components used in this step are shown. Complete the pages in order.



UNO GND is already on the − rail from Step 1; ESP32 GND is added again here for the UART link.

Vout = 5 × (1 k + 220) / (1 k + 1 k + 220) ≈ 2.75 V

R1 = 1 kΩ · R2 = 1 kΩ · R3 = 220 Ω. All three are required.

ESP32 GPIO17/TX2 → UNO D2 directly; ESP32 GPIO16/RX2 only through the divider.

The 10 kΩ resistor is spare for this build; it is not part of the divider.

CHECK BEFORE MOVING ON

- UNO D3 → R1 input (column 3). Divider node at column 6 → ESP32 GPIO16/RX2.
- ESP32 GPIO17/TX2 → UNO D2 directly.
- Divider GND at column 12 and ESP32 GND both reach the − rail.
- Never connect UNO 5 V directly to an ESP32 GPIO.